

Om Patel

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EDUCATION

California State University, Long Beach
Bachelor of Science in Computer Science

Jan 2024 – May 2027
Long Beach, CA

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

California State University, Long Beach

Long Beach, CA

- Designed and validated a binary deepfake detection classifier using CNN and transformer-based feature extraction, achieving **89% accuracy** and **0.91 F1-score** across synthetic media spanning **5 generation techniques**.
- Developed an automated MRI analysis pipeline using CNN-based semantic segmentation to classify osteoarthritis severity, reducing per-scan radiologist review time by **65%** on a labeled dataset of **600+ knee scans**.
- Standardized a shared Python-based data ingestion, augmentation, and model evaluation framework across both research tracks, cutting experiment setup time by **40%** and ensuring reproducible results across all model iterations.

Software Engineer, Student Assistant

July 2025 – Present

Beach Building Services (CSULB)

Long Beach, CA

- Integrated Gemini LLM into a production web application with structured prompt pipelines, input validation, and rate-aware fallbacks, sustaining **100+ concurrent requests/min** with **zero downtime** incidents.
- Modeled a normalized relational PostgreSQL schema for professor and course rating data across **50+ records**, designing analytics views, indexed lookups, and aggregation queries for stakeholder reporting.
- Optimized performance-critical C/C++ data parsing and validation utilities on Linux, cutting processing time by **30%** through memory-efficient algorithms and low-overhead I/O strategies.
- Shipped onboarding guided-flow features using component-driven frontend architecture, increasing feature adoption by **18%** measured via event tracking and funnel analytics.

AI Engineer Intern

September 2025 – December 2025

Impacter AI

Remote

- Built and deployed Python-based REST API microservices with ETL-style data transformation pipelines integrating LLM inference into a production web application, reducing end-to-end latency by **180ms** via async handling and response caching.
- Diagnosed concurrent request race conditions and improved LLM output accuracy by **12%** via structured output schemas, post-processing validation, and systematic prompt engineering.
- Fine-tuned GPT-3 and Llama 2 using few-shot prompt engineering and quantization strategies, cutting compute costs by **35%** and average token usage by **28%** across production workloads.

PROJECTS

RepoSage AI | *Python, FastAPI, LangChain, PostgreSQL, pgvector, OpenAI API, GitHub API, Redis, BullMQ, Docker*

May 2026

- Engineered an agentic RAG system over GitHub repositories using LangChain tool-calling, enabling multi-step code understanding, architecture explanation, and bug detection across **50k+ line** codebases.
- Orchestrated an async distributed embedding pipeline with BullMQ and pgvector, applying semantic chunking strategies to improve retrieval precision by **38%** over naive fixed-size approaches.
- Generated structured onboarding docs, refactor suggestions, and architectural summaries via chained LLM prompts grounded in retrieved context, surfacing actionable insights in **under 5 seconds**.

DataFlow | *Python, Airflow, PostgreSQL, dbt, Redis, FastAPI, Docker*

March 2026

- Architected a multi-source ETL pipeline ingesting data from **6+ APIs** (Shopify, Stripe, Amazon SP-API) in Python, normalizing disparate schemas into a unified PostgreSQL warehouse processing **500k+ records/day**.
- Automated Airflow DAGs for scheduled incremental syncs with idempotent deduplication, retry logic, and dead-letter queuing, achieving **99.5% pipeline reliability** across all source connectors.
- Enforced data quality via dbt schema tests, null-check assertions, and row-count anomaly alerts, reducing malformed records reaching the analytics layer by **78%**.

PulseLink | *Go, Redis, PostgreSQL, Docker, Kubernetes, Kafka, Prometheus, Grafana*

January 2026

- Constructed a high-throughput URL shortener in Go handling **10k+ req/sec** with a Redis caching layer that slashed PostgreSQL read load by **85%** under peak traffic conditions.
- Deployed on Kubernetes with auto-scaling and health probes; wired Kafka for async event publishing and implemented adaptive rate limiting with exponential backoff to sustain **99.9% uptime**.
- Instrumented end-to-end observability with Prometheus and Grafana, surfacing **p50/p99 latency**, cache hit rates, and consumer lag for real-time performance visibility.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, Go, Java, JavaScript, TypeScript

AI/ML & Data: PyTorch, LangChain, RAG, Embeddings, Agents, Fine-tuning, Prompt Engineering, Airflow, dbt, Kafka, Spark, ETL/ELT, pandas, NumPy, OpenCV, MLflow

Backend & Storage: FastAPI, Node.js, Flask, PostgreSQL, pgvector, BigQuery, MySQL, MongoDB, Redis, AWS, Google Cloud, Firebase

Systems & Tools: Linux, Git, Docker, Kubernetes, CI/CD, GitHub Actions, REST APIs, WebSockets, Prometheus, Grafana, LangSmith, Agile